



Scientific investigation & logbook

The methodologies, the role of the logbook, primary vs secondary data, and the prescribed scientific-poster format used to report your student-designed investigation.

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◆ What Outcome 3 actually asks

Across Units 3 & 4 you design and conduct one student-designed scientific investigation related to biodiversity, environmental management, climate change and/or energy, then report it as a scientific poster supported by a logbook. Investigation-design and analysis skills are also examinable in the end-of-year exam.

The spine of the task

Aim & question → hypothesis → chosen methodology → method (with controlled variables) → generate primary data → analyse for patterns, error & uncertainty → evidence-based conclusion → evaluate limitations → communicate on a poster.

1 The scientific methodologies

Pick the methodology that fits the question. Examiners reward students who can name it and justify the choice.

Controlled experiment

Manipulate an IV, control other variables, measure the DV. The only methodology that can establish cause.

Correlational study

Measure two variables as they naturally occur to see if they co-vary. Shows association, not causation.

Fieldwork

Observe and measure in the natural environment — quadrats, transects, water sampling, surveys.

Modelling

Build a physical, conceptual or mathematical representation to explore a system (e.g. an energy-balance model).

Simulation

Run a computer or physical model through scenarios to predict behaviour where real testing is impractical.

Literature review

Synthesise existing secondary data to answer a question or build a case.

Classification & identification

Arrange or name organisms, materials or phenomena by shared characteristics.

CORRELATION $\neq$ CAUSATION

A correlational study can show two things rise together; only a controlled experiment, holding other variables constant, can argue one causes the other.

2 Method & variables

Element	What examiners look for
Independent variable	The single factor changed, with its range and units.
Dependent variable	What is measured, operationalised with units & instrument.
Controlled variables	Named factors held constant, each with <i>how</i> it's kept constant.
Repetition	Trials/replicates to manage random error and assess reliability.

3 Primary vs secondary data

Primary data

Generated **first-hand** by you in your own investigation — your quadrat counts, your temperature logs. Authenticated by your logbook.

Secondary data

Generated by **others** — BoM records, agency datasets, published studies. Useful for context, calibration and comparison; must be referenced.

4 The logbook — how work is authenticated

The logbook is the evidence trail that the data is genuinely yours.

What belongs in it

Dated entries as you work: the evolving method, raw data with units and conditions, calculations, observations, photos, changes and the reasons for them, and references. Continuous, contemporaneous and unedited — a reconstructed-after-the-fact logbook is the classic red flag.

AUTHENTICATION CHAIN

Every figure on the poster should trace back to a raw entry in the logbook. That traceability is what makes the result defensible.

5 The scientific poster

VCAA prescribes the structure. Learn the parts and the word budget.

Section	Contains
Title	Concise statement of the investigation; student name(s).
Introduction	Context, the scientific concepts, the question/aim and hypothesis.
Methodology & method	The methodology chosen and a summary of the method, including variables.
Results	Processed data as appropriate tables/graphs — not raw logbook dumps.
Discussion	Interpretation, links to the concepts, error/uncertainty, limitations.
Conclusion	An evidence-based answer to the question; hypothesis supported or refuted.
References & acknowledgements	Sources of secondary data and any assistance.

THE CENTRAL STATEMENT

The poster carries a single communication statement of ~600 words max reporting the findings — the heart of the poster, written for a scientific reader.

! Exam traps

“It’s reliable because it’s accurate”

Different ideas. Reliability = consistency on repetition; accuracy = closeness to the true value.

Controlled variable ≠ control group

One is a factor held constant; the other is an untreated comparison group.

Conclusion overreach

Don’t claim cause from a correlational design, or generalise beyond the sample.

Raw data on the poster

The poster shows *processed* results; raw data stays in the logbook.

★ Self-check

A student wants to know whether shade lowers soil temperature. Identify the most appropriate methodology and justify it. (3 marks)

Weak: They should do an experiment because experiments are best.

Strong: A controlled experiment is most appropriate: by manipulating the independent variable (shade present vs absent) while holding other variables constant (soil type, time, depth), the student can attribute any difference in soil temperature to shade — establishing cause, which a correlational study could not.

Explain how a logbook authenticates a student's primary data. (2 marks)

Weak: It shows they did the work because it's written down.

Strong: The logbook records dated, contemporaneous raw data, method changes and observations as the investigation proceeds, so each result on the poster can be traced back to an original entry. This continuous trail makes it possible to verify the data was genuinely generated by the student.



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